

MA 374: Financial Engineering Lab

Lab Assignment 01

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January 9, 2026

Problem Description

We consider a standard Binomial Model for pricing European Call and Put options with the following parameters:

- Initial stock price: $S(0) = 9$
- Strike price: $K = 10$
- Time to maturity: $T = 3$ years
- Risk-free interest rate: $r = 0.06$
- Volatility: $\sigma = 0.30$

The binomial model parameters are computed using continuous compounding:

$$\begin{aligned}\Delta t &= \frac{T}{M}, \\ u &= e^{\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t}, \\ d &= e^{-\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t}, \\ \tilde{p} &= \frac{e^{r\Delta t} - d}{u - d}.\end{aligned}$$

For each value of M , the no-arbitrage condition

$$d < e^{r\Delta t} < u$$

was verified and found to hold.

Part (A): Initial Option Prices for Varying M

For different values of the number of time steps M , the initial prices of the European Call and Put options were computed. In each case, the no-arbitrage condition was confirmed.

Table 1: Option prices at $t = 0$ for various values of M

Steps (M)	Call Price C_0	Put Price P_0
1	2.46336686	1.81606897
5	2.18324492	1.53594704
10	2.12483539	1.47753751
20	2.12958331	1.48228543
50	2.12721517	1.47991728
100	2.12366674	1.47636886
200	2.12020950	1.47291161

Part (B): Convergence of Option Prices

The initial option prices were plotted against the number of time steps M . Two plots were generated:

- Prices computed for every integer value of M
- Prices computed at intervals of 5

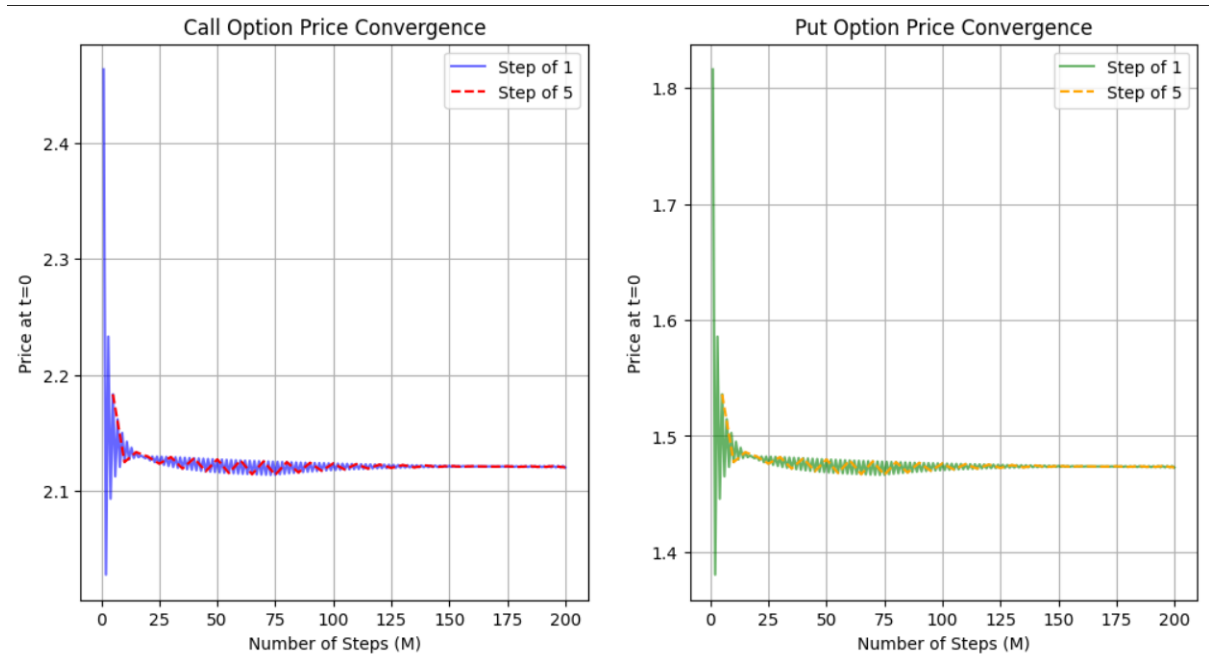


Figure 1: Convergence of Call (left) and Put (right) option prices with increasing M

Observations

1. The option prices exhibit oscillatory (sawtooth-like) convergence rather than monotonic convergence.
2. The amplitude of oscillations decreases as M increases, indicating convergence toward the continuous-time (Black–Scholes) price.
3. Sampling M in steps of 5 skips intermediate oscillations and may visually appear smoother.

Part (C): Option Values at Intermediate Times ($M = 20$)

For $M = 20$, option values were computed at selected intermediate times:

$$t = 0, 0.30, 0.75, 1.50, 2.70.$$

The no-arbitrage condition was again verified and satisfied.

European Call Option Values

Table 2: European Call option values at different nodes

Time t	Option Values
0.00	2.12958331
0.30	3.72475024, 1.99916798, 0.94923909
0.75	7.74065626, 4.67517654, 2.54429843, 1.21102726, 0.48772601, 0.15996339
1.50	20.27958825, 14.18652208, 9.39365148, 5.71701249, 3.07175858, 1.38553229, 0.49339033, 0.12793143, 0.02122405, 0.00167743, 0.00000000
2.70	66.05705718, 50.32328563, 37.85198001, 27.96665382, 20.13109298, 13.92026975, 8.99728771, 5.09510740, 2.14584853, 0.46099915, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000

European Put Option Values

Table 3: European Put option values at different nodes

Time t	Option Values
0.00	1.48228543
0.30	0.82357237, 1.46298877, 2.28766577
0.75	0.20620200, 0.51471205, 1.05821226, 1.84477543, 2.80175167, 3.80585369
1.50	0.00052210, 0.00748449, 0.04977501, 0.20570547, 0.59832106, 1.32004884, 2.33656120, 3.48398885, 4.57646441, 5.50744495, 6.25919873
2.70	0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.14378739, 0.91062768, 2.39294970, 3.93331477, 5.15427845, 6.12207000, 6.88918577, 7.49723675, 7.97920571, 8.36123632, 8.66405122